

1. Consider a simple regression model $y_i = \beta_0 + \beta_1 x_i + u_i$. Let $\{(x_i, y_i): i = 1, \dots, n\}$ denote a random sample of size n from the population.

(a) Prove that given the data $\{(x_i, y_i): i = 1, \dots, n\}$, $\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ minimize $\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$.

(b) Suppose the zero conditional mean assumptions holds. Prove that $\hat{\beta}_1$ and $\hat{\beta}_0$ estimators are unbiased. Make sure every step of your proof **is clearly explained and justified**.

(c) Derive $Var(\hat{\beta}_1|x)$

2. Explain what factors affect the magnitude of $Var(\hat{\beta}_1|x)$. In practice, how would you decrease the $Var(\hat{\beta}_1|x)$? [Note: No need to submit the answer for this question.]